Chapter Three. Fitting

Staunch Systems Trader

This entire chapter is about using data to create the **trading rules** used by **staunch systematic traders**. It is not necessary reading if you are going to use my **framework** to make discretionary forecasts as a **semi-automatic trader** or without any rules at all as an **asset allocating investor**.

A DECISION TO RUN A SYSTEMATIC TRADING SYSTEM MEANS YOU have to select one or more **trading rules** and discard others as being unworthy. This process is often called **fitting**. Given our human tendency to be overconfident this procedure is fraught with danger. You need to beware of **over-fitting**:[40] selecting a set of trading rules that fit past data too well and which are unlikely to make money in the future.

Chapter overview

The perils of over-fitting	The dangers of trying to selectively choose and fit rules based on past data.
Effective fitting	Steps you must follow if you insist on fitting your trading rules.
How I choose my rules	The process I use to avoid fitting almost entirely.

40. Another common term for this is **curve fitting**.

The perils of over-fitting

The 50 model kid

Shortly after leaving the hedge fund industry, I began discussions about consulting, and managing some capital, for Aqueduct Capital,[41] a local proprietary trading firm. The office contained the usual mixture of grizzled ex-LIFFE traders and naïve youths, all day trading a handful of futures contracts. But the boss was particularly proud of his quantitative team which consisted of a couple of 20-somethings toting PCs running an off-the-shelf **back-testing** software package.

"This is Joe. He's only been here a month and he's already come up with 50 new trading rules that are profitable in back-tests!" exclaimed the boss.

"Yes, this software is amazing. It can automatically test hundreds of rules a day," added Joe.

I managed to keep a straight face and replied as diplomatically as I could, "Well I am sure some of them will work."

Inevitably the joint venture discussions then broke down, which was fortunate as the firm was liquidated a few months later. The discovery of apparently profitable rules sifted from thousands of possibilities is an incredibly dangerous approach, for reasons that will become apparent in the rest of the chapter.

Ideas first testing for rules and variations

Before understanding why Joe was on the wrong path you need to be clear on what **fitting** actually involves. I'm going to restrict my attention in this chapter to the **ideas first** method that I introduced in chapter two. You already know that I prefer the ideas first approach, but the reason I use it here is because it's much easier to explain and understand how to avoid over-fitting, than with the alternative of **data first**.

The fitting process is going to involve selecting one or more **trading rules** from a list of candidates, each based on a brilliant idea. Let's look at a contrived example, but beware this is not a rule I would recommend using. The basic hypothesis is that the British pound/US dollar currency rate seems to move in ranges as figure 6 shows. You might think that buying pounds if they're 5% below their average over the last year, and selling at 5% above, is a good strategy. This is an example of a **mean reversion** rule.

You would now test this initial rule on historical data and look at how it performs and behaves. At this stage if the rule is unpromising you can drop it and move on to the next idea. If you like it enough you can proceed to the next stage which I call **calibration**, although it's possible and often desirable to just stick with the initial version.

41. All names have been changed to protect the ignorant.

FIGURE 6: GBP VS USD EXCHANGE RATE

During calibration you examine some **variations** on the basic trading rule. In the simple example you could test the original range of plus or minus 5% against alternatives such as 3%, 6% or 10%; or you can compare the current price against the average price over the last year, or week, or two years and so on. Usually you would choose the most *profitable* rule using a performance measure like the **Sharpe ratio**. Calibration can also be used – as I will show later in the book – to find rules which *behave* in a given way, such as trading at a given speed. You can then decide which variation, or variations, to keep.

Once you have chosen a portfolio of trading rules and variations you need to decide how to allocate your capital amongst them. Portfolio allocation decisions like this are the subject of the next chapter. For the moment it's worth noting that a poor variation can be rejected out of hand, or given a relatively small allocation in the overall system.

What if you want to use data first methods? Then you will need to be sufficiently expert to apply the principles in this chapter with your own preferred tools.[42] You should not use a data first method for which you don't have any deep understanding. I strongly suggest that you do not use a method blindly, just because it came packaged with some **back-testing** software.

42. The main danger of the data first process is that you allow too many degrees of freedom. (I will not be defining this term since anyone using a data first approach ought to understand it already. If you don't you're in trouble!) Whereas the danger of ideas first is that you test too many ideas, or variations of ideas, until you find one (or 50!) that work. Either approach can result in over-fitted trading rules which look great in **back-test** but underperform once actually trading.

Before we continue here's a final note of caution about ideas first testing. Any worthy trading systems designer will know their history, have read all the right textbooks and be aware of what other people trade. You could end up testing only ideas that you already know will work, which is a form of implicit **over-fitting**. As a result no matter how careful you are with the subsequent fitting process your back-tested **Sharpe ratios** will probably still be overstated – beware overconfidence!

Cheating with a time machine

Let's consider the first common mistake when fitting, which is pretending that you had access to a time machine.

When fitting there are always two distinct time periods. First there is the period used to *fit* the model, and secondly the time period used to *test* it. To illustrate this consider again the example of trying to fit a GBPUSD trading rule, and assume you're trying to find the best single variation. Suppose you've got ten years of daily price data available to fit on.[43]

The easiest and probably most common method is to use the whole ten years as your fitting period, and find the single most profitable variation over that decade. You then go back and test how this single variation did over each of the same ten years. For each year that you are testing in, you use the same model, based on the entire ten years of data.

Figure 7 shows this for an example model running between 1990 and 2000. Each stage of the fitting occupies a row. So the first row shows that you would use all the data from the years 1990-2000 to fit the model which you test in 1990. In stage two you use the same fitted model to test performance in 1991, and so on.

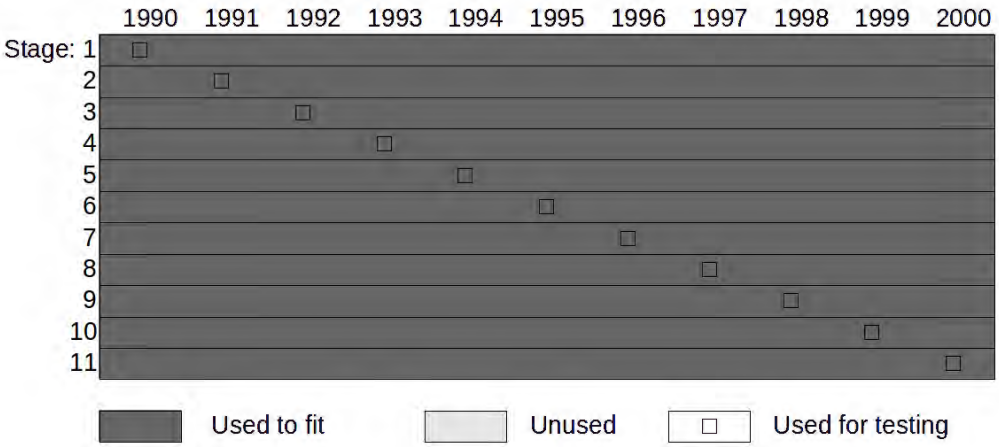
The performance of this **back-test** will be amazing; too good to be true. Since no time machine was really available you didn't actually have the whole ten years of data in 1990 and you wouldn't necessarily have chosen the right variation. This is known as an **in sample** back-test because you are using the same data to fit and test performance. In sample testing should be avoided as it will produce extremely optimistic back-test results, and favour more complex rules that fit the data better but won't do as well in real trading.

There are better alternatives. A common one is to split the sample into two historical periods as in figure 8. You take all the data from the first half, 1990 to 1994, and fit the best variation over that period. Then you use that single variation to test your performance on the second **out of sample** period, which in this case is each year from 1995 to 2000.

43. Throughout this book I'm going to assume that you are using daily data. This often means you can get longer histories of prices and it is appropriate for the kinds of trading rules I'll be discussing. Having data at a higher frequency is only necessary if you're testing very fast trading rules, such as those that expect to hold positions for less than a week.

The problem with this is that you waste half your data, and end up with only six years of performance to look at.[44] You also don't account for any potential changes in market structure in the second half of the data.

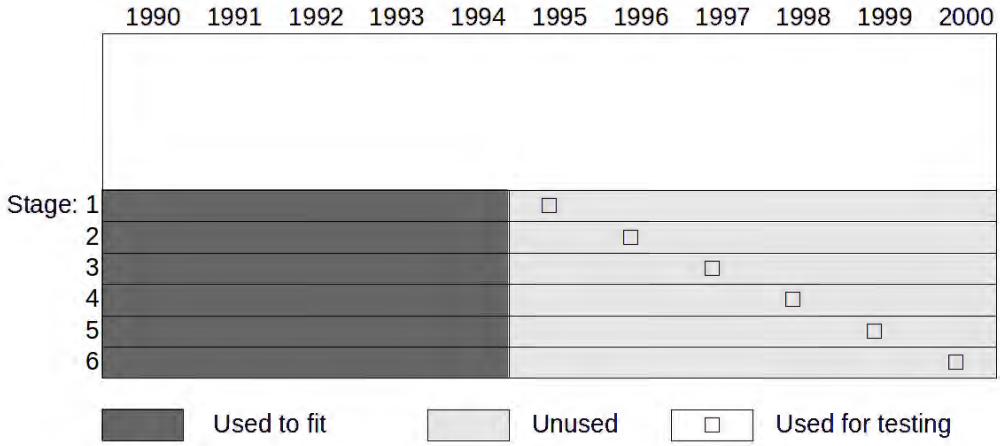
FIGURE 7: IN SAMPLE BACK-TESTING: EFFICIENT BUT DISHONEST



In sample testing. You use all the data to fit and then test performance from the start. In each stage (row) you test data for a different year, but use data from all years to fit.

44. You can also fit on the first half of the data and test on the second half as before, and subsequently fit on the second half and test on the first. This keeps more data but isn't entirely honest as it still requires a time machine, albeit one you use only once. Yet another variation is to use the entire dataset to fit except for the year you are testing. This also strikes me as relatively dishonest.

FIGURE 8: HALF OUT OF SAMPLE TESTING: HONEST, BUT WASTEFUL



You fit your trading rules on the first half, and then test performance on the second half. In each stage (row) you test a different year, but always using the first half of the data to fit. No testing is done in the first half of the data. The second half of the data isn't used for fitting.

My preferred solution is to use an **expanding** window, as in figure 9.[45] Suppose you think you need at least a year to fit your system. In the next stage you fit on the first year of data for 1990, and then test the resulting variation in the second year, 1991. Then in stage three of figure 9 you test in 1992 using the variation fitted with data from the years 1990 and 1991. To test 1993 you use the best variation fitted using 1990, 1991 and 1992.

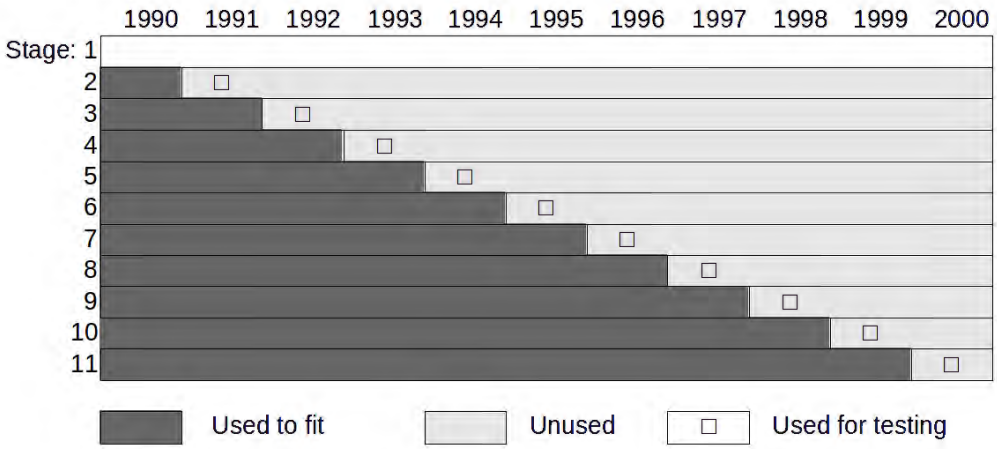
This continues until 2000 when you're using the previous ten years of data to select the best variation. Because you only fit using the past you're not cheating. You also use as much of the past as you 'legally' can, so nothing is wasted.

The only problem with this method is that if the world changes you'll still be using potentially irrelevant past data. To avoid this you could fit on the last five or ten years of data, once you have enough history to do so, and discard earlier years. This is a **rolling window**.[46] As figure 10 shows a five-year rolling window is identical to the expanding window until 1996. At this point you would discard 1990, and use only the years 1991 to 1995 to fit the best variation.

The length of the window needs to be short enough to pick up changes in market structure, but long enough to give statistically significant results. As you'll see later you often require multiple decades of data to fit models, which makes the use of rolling windows problematic.

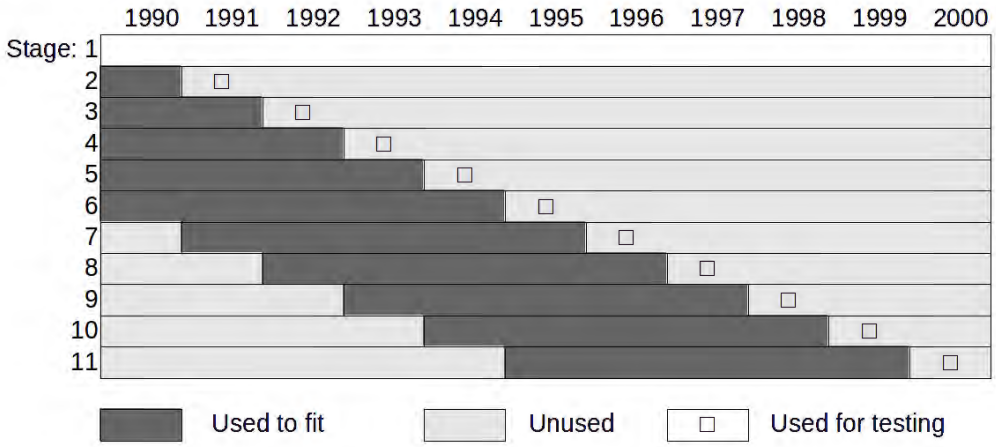
45. This is sometimes called *anchored fitting*.
46. Another widely used term for this is *walk forward fitting*.

FIGURE 9: EXPANDING OUT OF SAMPLE IS BOTH EFFICIENT AND HONEST



Every year you fit the data only on the past. Each row shows the fitting and testing done for a particular year.

FIGURE 10: ROLLING OUT OF SAMPLE ADAPTS TO CHANGING CONDITIONS, BUT MAKE SURE YOU HAVE ENOUGH DATA FOR STATISTICAL SIGNIFICANCE



Every year you use the past data to fit; up to five years' worth. Each row shows the fitting and testing done for a particular year.

When fitting goes bad

Now let's examine a concrete example of poor fitting practice. I'm going to select the best **variation** of the early loss taker **trading rule** I introduced in chapter one to run over the CME gold futures contract, out of a possible menu of 90 variations.[47] This might seem excessive, but it's fewer variations than Joe was testing in my earlier anecdote. Bear in mind that if I had a five parameter trading rule, and I allowed each parameter to take one of ten discrete values, then I'd be testing 10,000 variations!

Once I have at least one year of data I find the highest performing variation based on the **Sharpe ratio (SR)** from the previous 12 months. I then test the performance for the following year, again using the last 12 months of data. This process is then repeated annually. You should recognise this as a one year **rolling window back-test**.

Which of the following alternatives do you think will give me the best performance?

1. Picking the *best* variation: Each year I use the best performing variation from the previous year.
2. Picking a *random* variation: Ignoring my fitting, each year I choose one variation at random. Because the randomness of each choice will influence the results I run this experiment a number of times, and take the average performance.
3. Keeping *all* the variations: Again ignoring my fitting, I keep all 90 variations, and use an equally weighted average of their forecasts.

If you're a big fan of fitting the results are disappointing. Choosing the best rule each year from the previous year gives me a rather poor SR of 0.07. If I go for the second option and just choose a random rule annually each January 1st, then on average I get a Sharpe of 0.2. The best option is to forget about selecting trading rule variations entirely and run an equal blend of them all. This gives an SR of 0.33, which is pretty good for one kind of trading rule run on a single **instrument**. These results could be a fluke but I get similar results on many different instruments and trading rules.

Why does fitting do so badly?

Firstly choosing just one variation to run at a time smacks of overconfidence. As you'll see shortly we rarely have enough evidence that one rule is definitely better than another. Secondly one year of data is wholly insufficient to decide which trading rule is best. I make this problem worse by fitting on the history of a single instrument, gold futures. Finally, like Joe, I am testing far too many variations.

47. For those who are interested this is based on the 'A and B' system defined in appendix B. Values for B (the stop loss value) will be the integers from 1 to 10; to capture different trend lengths. Because I only want to look at trend following rules (early loss takers) I iterate only over values of A (the profit target) that are equal to or larger than B: A = 1 × B, 1.5 × B, 2 × B, ... 5 × B. This gives 90 possible variations in all.

The hazards of rule selection

The multiple testing problem – even some random rules will look good

Why is it so dangerous to test large numbers of **trading rules** and **variations**? Apart from the effort involved it's very likely that you will end up picking a poor rule just by chance.

To see why here is an experiment. Let's suppose I am trying to achieve alchemy and find profitable rules where there are none to be had. I have a pool of a certain number of possible trading rules, all of which have true expected average returns of zero, although in a real scenario I wouldn't know this beforehand! The rules are entirely arbitrary and their returns are generated from random data.[48]

For each test I get one year of return data for each rule in my pool, as in the gold futures example above, and select all the rules whose **Sharpe ratio (SR)** in that year is higher than a given minimum level. If no rule has an SR above the threshold then I won't choose any. All the rules that pass the test will be kept (even if there are 50!).

Because the underlying data is random I need to generate new data multiple times and then repeat the test to get meaningful results. I then measure the average number of rules accepted given a specific minimum level and the size of the pool available. As table 3 shows, even if I'm strict and set an extremely high minimum SR of 2.0 I'll still pick up a couple of bad rules if I test enough of them.

TABLE 3: IF YOU TEST ENOUGH RULES, SOME BAD ONES WILL ALWAYS SLIP THROUGH

Number of rules tested in pool	Minimum Sharpe ratio		
	0.5	1.0	2.0
1	<1	<1	<1
5	1.4	<1	<1
10	3	1.5	<1
50	16	8	1.2
100	30	16	2.3

The table shows the average number of rules accepted from pools of arbitrary unprofitable rules, given different pool sizes (rows), which were tested to see if their Sharpe ratio exceeded a minimum cutoff (columns).

48. Many of the examples in this chapter will use imaginary daily returns of various arbitrary trading rule variations. These fake returns are randomly generated with the kind of characteristics I want: expected mean, standard deviation (from which we get the Sharpe ratio), and where relevant **correlation** with other variations. I then run these tests many times and report the average result, so the answer is not influenced by how each series of random numbers come out.

Practically then what SR cutoff should I use? As none of the imaginary rules are truly profitable ideally I wouldn't accept any. But in real fitting situations we can't set the bar too high or even good rules will be discarded; after all a realistic SR for a real trading rule, tested on one **instrument,** is only likely to be around 0.30. Let's suppose I would be happy with a 5% chance of picking out at least one rule that was truly unprofitable. Also in the real world I'd usually have more than one year of data, so let's see what effect additional history has on my findings.

Table 4 has all the results. As you might have suspected the length of the return series is very important here. More history means less chance of a zero SR rule getting a lucky streak, so I can set the cutoff lower. However, even with 30 years of data I can't risk testing more than a handful of rules, and even then the cutoff is far too high when many perfectly good variations will only have true Sharpe ratios of 0.3.

TABLE 4: WITH MORE HISTORY YOU CAN SET A LOWER SHARPE RATIO THRESHOLD TO AVOID PICKING A BAD RULE FROM A LARGER POOL

Number of rules tested	Years of data			
	1	5	10	30
1	1.5	0.7	0.5	0.4
5	2.3	1.1	0.8	0.5
10	2.8	1.2	0.8	0.6
50	3.4	1.5	1.0	0.6
100	3.4	1.5	1.1	0.7

The table shows the Sharpe ratio cutoff needed when testing a given size pool (rows) of trading rules, with a certain amount of years of historical data (columns) to ensure you only have a 5% chance of picking out one or more truly bad rules.

How much history do you need to decide if a rule is good?

You've now seen that a common mistake is to use insufficient historical data to select or calibrate a rule. How much data do you need? How long do you need to decide if a rule has a positive **Sharpe ratio (SR)** and is worth keeping?

To answer this I generated more random trading rule daily returns, this time assuming an underlying positive Sharpe ratio, which again I wouldn't know in advance. As more trading history is generated I can estimate the SR each year and the average so far. At the same time I look at the distribution of those annual Sharpe ratios.[49] This allows me to get

49. For a more technical discussion of this issue, see Andrew Lo's paper 'The Statistics of Sharpe ratios' in *Financial Analysts Journal* 58:4, July/August 2002.

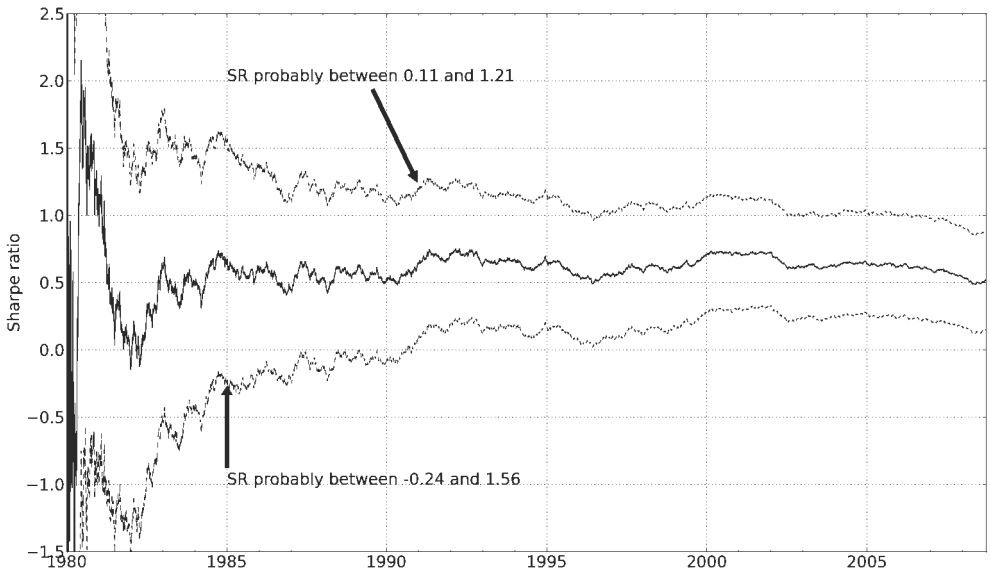
a feel for how statistically significant my measured average SR is; was I just lucky or is this really a good system?

A classic test for statistical significance is the T-Test. In this example it determines whether an estimated Sharpe ratio is likely to be positive given the estimate of its mean and **standard deviation**. The further away the mean is from zero, as measured in units of **sigma**, the more likely the unknown SR is actually positive.

This test is commonly used with a threshold of two **sigma**. If an estimated average SR is more than two standard deviations above zero there would only be a 2.5% chance of this happening if the true SR was actually negative.

Figure 11 shows the evolution of the average measured SR for an arbitrary trading rule, and around it the upper and lower 'confidence intervals'. Each confidence interval is two **sigma** away from the average, so when the lower interval pushes above zero I know there is only a 2.5% chance the trading system is really a loss making rule in disguise.

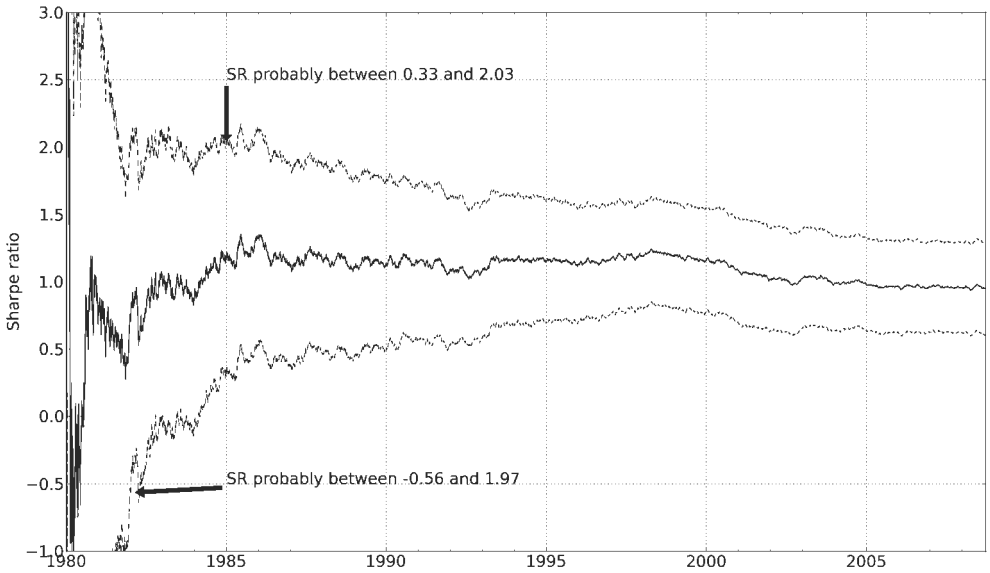
FIGURE 11: WITH A TRUE SHARPE RATIO (SR) OF 0.5, IT TAKES MORE THAN TEN YEARS TO PASS THE T-TEST AND CONCLUDE THE SR IS PROBABLY POSITIVE



In the figure you can see the average SR converging quickly on the true value of 0.5 SR. But it takes over ten years before the lower confidence interval goes above zero and the T-Test is finally passed. Only then can we be reasonably certain this is not a loss making system.

If I repeat this for a rule with a true SR of 1.0 I get figure 12. The time to pass the test here is just a few years. On average it will always take less time for higher SR strategies to prove their profitability.

FIGURE 12: WITH A TRUE SHARPE RATIO OF 1.0 WE PASS THE T-TEST IN A FEW YEARS



Let's find the average time to pass the test for a given true Sharpe ratio. After running the necessary experiments I get table 5. Except for very good rules you need at least ten years, and usually more, to be sure a strategy makes money.[50] The average trading rule for one **instrument** has a realistic SR of around 0.3; so you'd need nearly 40 years of history!

TABLE 5: IT TAKES DECADES OF DATA TO SEE IF MOST STRATEGIES ARE LIKELY TO BE PROFITABLE

True Sharpe ratio	0.2	0.3	0.4	0.5	0.7	1.0	1.5	2.0
Average years to pass profit T-Test	45	37	33	20	10	6	3	1.4

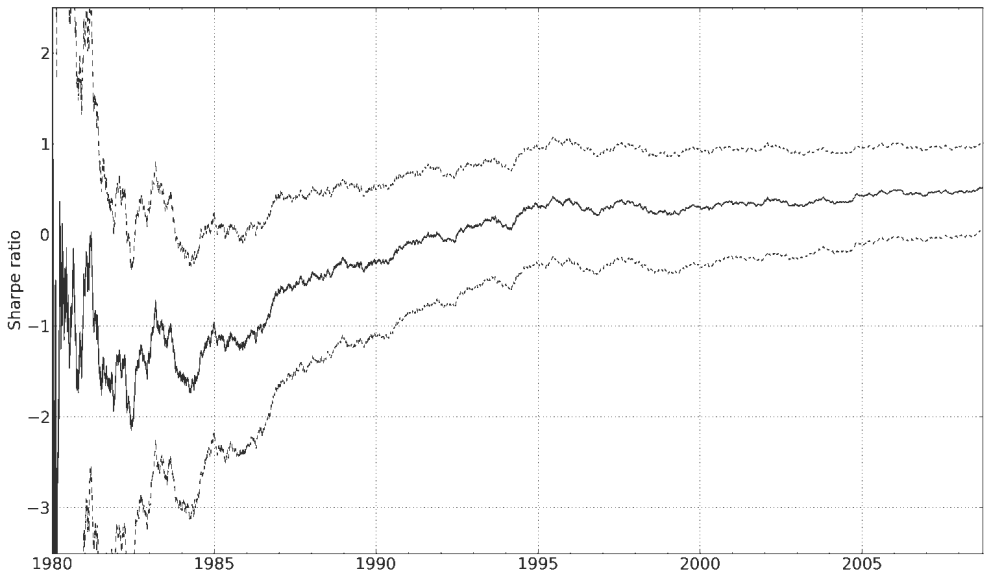
The table shows average time in years to pass the T-Test for profitability given the true Sharpe ratio of the trading rule.

50. These results also depend on the **skew** of returns. For example, with a true SR of 1.0 I needed about three years' more data for a typical negative skew volatility selling strategy to pass the T-Test, compared to a positive skew **trend following** strategy.

How much history do you need to decide if one rule is better than another?

Now what if you have some alternative rules and want to find the best. Let's suppose I again have two random rules, one with a true **Sharpe ratio** (SR) of 0.30, and the other an SR of 0.80. Initially I'm going to assume these are quite different rules with no **correlation** between their returns. This time to perform the T-Test I need to estimate the difference in the two Sharpe ratios, and measure the average and standard deviation of that difference.

FIGURE 13: IT TAKES THREE DECADES TO DISCOVER THAT THIS SR 0.8 STRATEGY IS PROBABLY BETTER THAN AN UNCORRELATED STRATEGY WITH SR 0.3



In figure 13 is the average difference in SR and the relevant confidence intervals for this difference. Once the lower confidence interval goes above zero then we can be reasonably certain that one rule is better, as again there is only a 2.5% probability of this happening by chance. You can see from figure 13 that the time to certainty is around 30 years!

Table 6 shows the average time taken to pass the T-Test when comparing an SR 0.30 rule with one that truly has a higher SR. These results vary depending on the difference in SR and the correlation of the rules. More closely related rules will be easier and quicker to distinguish for a given SR difference.[51] In practice distinguishing rules requires

51. More subtly the overall level of the Sharpes involved will also influence the results, as Andrew Lo's paper discusses. The skew of the rules is also important, even when comparing rules with similar skew.

considerable historical data except for the rare cases of variations which are both highly correlated, and also perform very differently.

TABLE 6: CAN YOU DISTINGUISH TRADING RULES GIVEN A FEW YEARS OF DATA? ONLY IF THEY ARE HIGHLY CORRELATED AND ONE HEAVILY OUTPERFORMS THE OTHER

Sharpe ratio advantage	Correlation between rules				
	-1.0	0.0	0.5	0.8	0.95
0.1	47	47	46	44	37
0.25	46	45	40	32	10
0.5	41	37	25	10	3
1.0	23	11	8	2.5	0.5

The table shows the number of years needed for us to be reasonably certain that one rule is better than another, given the SR advantage of the better rule (rows) and the correlation between the trading returns of the two rules (columns).

Should you fit one trading rule for all instruments?

In 2010 the large systematic hedge fund in which I worked was reorganised into **asset classes**. My team managed the fixed income portfolio, another ran equities and so on. Initially the trading strategies we ran were fairly similar. However over time, as we did more market specific research, we all became convinced that we needed to customise our systems differently.

In many cases we went further and also began to fit different parts of our portfolios separately. We might for example have fitted emerging and developed market bond futures with different trading rule variations.[52] Fitting separately is common, and many system traders take this to the extreme and fit every **instrument** separately. Almost all **back-testing** packages, such as the one Joe was using, default to fitting trading rules on a single instrument at a time. So you would have one **variation** of a rule for the British pound/dollar, another for euro/dollar and so on.

This is a classic example of the narrative fallacy, a **cognitive bias** we have seen before. To our human minds it makes sense to have different stories, and so different trading rules, for different instruments.

52. In reality we didn't do this. Or did we? Unfortunately I can't give real examples without breaking the terms of my non-disclosure agreement.

But is this the right thing to do? It depends on whether the behaviour of each variation over the various instruments is significantly different. If it isn't then you're likely to be **over-fitting** by individually tailoring each rule for every instrument. In practice there isn't usually a statistical difference between different instruments, particularly if they are closely related such as a group of equity index futures.[53]

In fact pooling your data and fitting multiple instruments together is an excellent way of getting more data history, which gives you a better shot at being able to distinguish profitable and loss making trading rules.

As an example I have two trading rules in my own portfolio which for a typical single instrument have **Sharpe ratios (SR)** averaging 0.05 and 0.30 respectively. Despite being perfectly uncorrelated, table 6 shows I'd need 45 years to distinguish them; almost impossible when even the earliest financial futures only began trading in 1972. However the same rules run on a portfolio of instruments have SR of 0.13 and 1.13.[54] The table shows I'd need just 11 years of data to distinguish these two Sharpe ratios from each other.

To make pooling easier you should design trading rules that are generic and can work with any instrument. I'll give some examples of generic rules in chapter seven, 'Forecasts'.

Some rules for effective fitting

If you haven't been warned off fitting, and insist on using data to select and calibrate trading rules, then following this advice should keep you out of serious trouble. Those who are now rightly terrified of fitting can skip ahead to the next section, where I show you how I avoid fitting almost completely.

Keep it simple

I am not a fan of complex fitting methods and I prefer not to use them. But if you do consider yourself an expert in a dark statistical art then naturally you would want to practise it. Just be aware; the more complex the method the harder it will be to realise when you are **over-fitting**.

53. There are some exceptions, notably when there are different trading costs for the instruments concerned, which I'll cover in chapter twelve, 'Speed and Size'.

54. Technical note: These are the returns you'd get if you traded an equally weighted portfolio of instruments using the same trading rule. The improvement in **Sharpe ratio (SR)** comes from the diversification effect across instruments. The alternative method is to stitch consecutive series of instrument returns together. This gives a very long history of returns, but the SR of the stitched series will be almost the same as the average across individual instruments (it is not identical to the average because of the effect of varying lengths of price series for different instruments).

Fewer alternatives

As table 4 shows reducing the pool of **trading rules** and **variations** considered is crucial unless you have many years of data.[55]

Ban time machines

You should always use **rolling** or **expanding out of sample** fitting. If using a **rolling** window don't make it too short; be aware of the number of years needed to distinguish rules from random noise, and each other (tables 4, 5 and 6).

Don't drop rules casually

Before picking one rule or variation and discarding others based on past performance, look carefully at table 6. It's unusual in reality to find highly **correlated** rules with radically different performance. More likely you will find very different rules with perhaps a **Sharpe ratio (SR)** difference of 0.5 but almost no correlation, or highly correlated rule variations which are very similar in SR. In both cases you need 30 years of data to differentiate them.

It's rare to have 30 years or more of price history, so it's difficult to justify picking one rule over another on performance alone.

Pool data across instruments

You're going to need all the data you can to make good fitting decisions. The easiest solution is to pool data from multiple **instruments**.

Only if there is a statistically significant difference in performance between various rules across instruments should you fit them individually. In practice this is rarely necessary.

Compare apples with apples

I've assumed throughout this chapter that you will use **Sharpe ratio (SR)** to compare rules. But comparing a positively **skewed** rule and a negatively skewed alternative purely on SR is highly misleading, as negative skew rules will tend to have flatteringly high SR.

When the future won't be like the past

Be careful of focusing on outright performance, rather than returns relative to benchmarks. This is very dangerous because many **asset classes**, like equities and bonds, have done extraordinarily well over the last 40 years or so. The easiest way to get extra **back-tested** profits is to use trading rules which are more highly **correlated** with the underlying asset class.

55. If you are using a more complex fitting technique then you need to keep your degrees of freedom appropriately small.

This is unlikely to work in the future as the main cause of these high returns was significant falls in inflation; something that won't be repeated.

How I choose my rules

There are many ways to navigate the fitting minefield. My own preference is to avoid it almost completely, by selecting trading rules and variations without looking at actual performance. I use the following process:

1. Come up with a small number of **trading rules** to exploit each idea I have about how the market behaves.
2. For each rule select a few **variations**. At this stage I am not looking at performance, but at behaviour such as trading speed and **correlation** with other variations.
3. Allocate **forecast weights** to each variation, taking uncertainty about **Sharpe ratios** into account. Poor rules will have lower weight, but are rarely entirely excluded.

This process means that I don't use performance to fit trading rules and variations. Instead returns data is reserved for finding forecast weights. These weights determine in what proportion each variation is used to predict each instrument's returns. The next chapter will explain how this portfolio allocation can reduce the weight on apparently poor variations without risking over-fitting.

Any variation that ends up with a negligible weight in the final portfolio can ultimately be dropped from your live trading system, reducing the complexity of implementation. This gives the same end result as excluding the rule earlier, but it means any **back-tested** performance incorporating the excluded rule will be more realistic.

Start with a small number of ideas

I prefer to come up with a relatively small number of ideas for trading rules. In my own portfolio I have eight rules drawn from five different themes, but if I was starting from scratch I'd begin with just a couple of rules: the **trend following** and **carry** rules that I will show you in chapter seven, 'Forecasts'.

You should have a *small* number of rules because:

1. Less risk of over-fitting: Fewer ideas means less chance of over-fitting, and a lower critical **Sharpe ratio** is required before accepting a rule (table 4).
2. Many ways to skin the feline: One kind of market behaviour can be picked up in multiple ways. For example we can capture trends with **momentum** oscillators, divergence and breakout systems to name but a few. There is little value in having dozens of similar rules for one kind of behaviour.

3. Keep it simple: With my process all the rules will survive the selection process so starting with too many rules will create complexity in the trading system. Complex systems are more difficult to trust and understand.

Far too much time and effort is spent by both amateur and professional trading system designers in looking for more, and better, trading rules. But the marginal value of adding additional rules is low, especially if they are of the same trading style. The average **correlation** between different rules trading a particular **instrument** is higher than between instruments trading the same rule. So diversification amongst instruments is preferable to rule diversification.

Adding new instruments is a tiresome task of uploading and checking data which is less fun than coming up with more trading rules, but in my experience is of far more benefit.

Keep the right number of variations

Although I prefer to have relatively few **trading rules** I wouldn't normally have just one variation of any rule which can be calibrated. But there is no need to use 90 or more possible combinations! For example if you're trying to capture price **momentum** then you will probably need a variation that captures relatively fast trends, one that captures very slow moves and perhaps two to three in between.

If two of your variations have more than a 95% **correlation** you can safely drop one of them since it will have almost no marginal benefit. You can also remove variations whose trading costs will be too high, or which trade extremely slowly and so are unlikely to give you significant returns. But do not drop variations purely because of their performance.

Don't look at returns - yet

My preference is to reserve actual historic performance data for deciding which **forecast weights** to give to **trading rules** and their **variations**. Done properly this can give you realistic **back-tested** performance and still down-weight poor trading rules. But if you've already used real data to pre-select only good rules then your back-test performance will be too optimistic, and you'll have **over-fitted**.

So at this stage it's only behaviour such as **correlation** and likely trading costs you should be looking at, not performance. You must avoid contaminating the back-test.

There isn't much detail about this process here but there will be examples of how it works later in the book. Whether you fit your trading rules, or use my hands off approach, the next problem is to decide how much of rule X and how much of Y – and how much of **instrument** A or B – to use in your trading system. This problem of **portfolio allocation** is covered in the next chapter.

Chapter Four. Portfolio Allocation

Staunch systems trader

Asset allocating investor

This chapter is about deciding how you share out your trading capital between different **instruments** or **trading rules**. Deciding the allocation between instruments is important for **asset allocating investors**, whilst **staunch systems traders** have to make both kinds of decision.

It isn't relevant if you're a **semi-automatic trader**, since you won't use systematic trading rules and will trade different instruments opportunistically. You can skip this chapter.

DECIDING HOW TO ALLOCATE WITHIN A PORTFOLIO OF ASSETS IS a problem every investor faces. How much in equities, bonds or cash? Should you split your equity allocation evenly between countries or just stick it all in the USA?

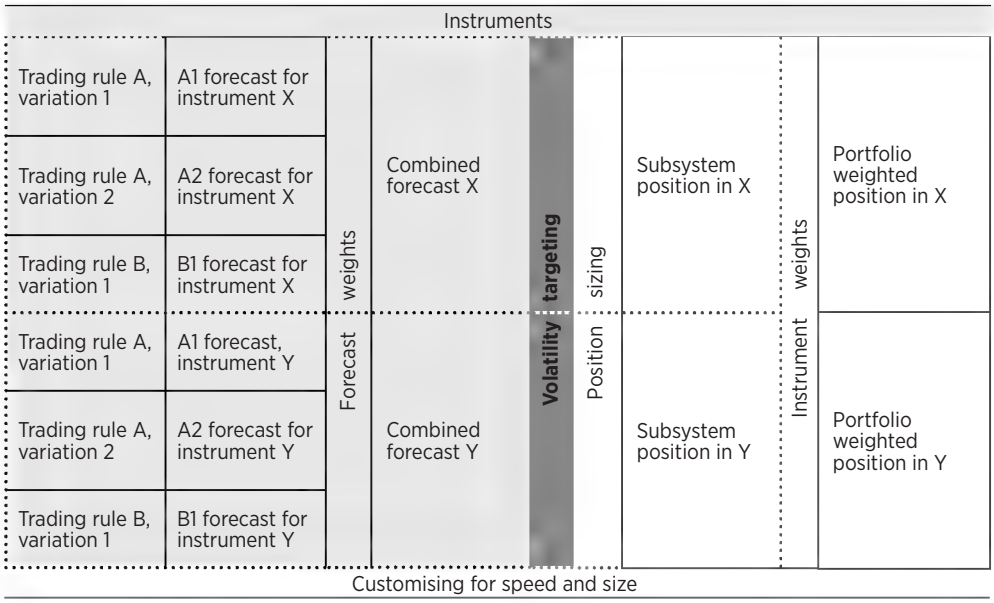
Allocation decisions are equally important for systematic investors and traders. If you're a **staunch systems trader** running more than **one trading rule**, including any **variations**, then you need to decide what **forecast weights** to use when you combine rules together to forecast the price of each instrument.

Both staunch systems traders and **asset allocating investors** also need to decide **instrument weights**; how much of your portfolio to put into the trading systems you have for each **instrument**. Because the tools in this chapter are for making both kinds of decision, I'll refer to portfolios of generic assets, which could be either instruments or trading rules. Later in the book I'll show you specific examples of each of the two types of allocation problem.

Just like trading rules, portfolio weights can be **over-fitted**. Optimising weights can give you a portfolio which does really well in **back-tests**, but which fails badly when traded

--- Chapter 9: Volatility Targeting (pp 152-170) ---

Chapter Nine. Volatility targeting



DO YOU REMEMBER SERGEI, DISPENSER OF OPAQUE TRADING advice in chapter seven? He told me that there were three issues to consider when deciding how big a position we should have. The first thing to consider was how much we liked the trade. You've dealt with that by creating a single forecast for each instrument you're trading – either a discretionary forecast for **semi-automated traders**, a constant forecast for **asset allocating investors**, or a **combined forecast** for **staunch systems traders**. Now we're ready to return to Sergei's second question, "How much can you afford to lose?"[96]

96. Many trading books use the term 'money management' to cover both volatility targeting and position sizing (which is discussed in the next chapter), but I prefer to separate them out for the reasons I explained in chapter five, 'Framework Overview'.

Chapter overview

The importance of risk targeting	Why getting your appetite for risk right is so important and the key issues involved.
Setting your volatility target	The measure of how much risk you're prepared to take and how to calculate it.
Rolling up profits and losses	How your volatility target should be adjusted when you lose or make money.
What percentage of capital per trade	Relating the volatility targeting concept to traditional money management where you limit bets to a specific percentage of capital.

The importance of risk targeting

Deciding your overall trading risk is the most important decision you will have to make when designing your trading system. Nearly all amateur traders lose money and most do so because their positions are too large compared to their account size.[97] Suffering painful losses is the main reason why both amateurs and professionals meddle with trading systems rather than letting them run unimpeded.

Making this decision correctly involves understanding two things. Firstly you must *understand your system*, in particular its likely performance and whether it's likely to have positive or negative **skew**. You must avoid **overconfidence**, as I discussed in chapter one. You should not extrapolate over-fitted **back-test** results to create high expectations of return. Are you sure your strategy can generate a **Sharpe ratio** of 2.0 after costs? Are you really, really sure?

Next you must *understand yourself*, in a complete and honest way. Can you face the possibility of regularly losing 5%, 10% or 20% of your own capital in a day?

These points also apply to those paid to manage other people's money. Additionally professionals need to understand their clients' tolerance for losing money. If investors aren't truly comfortable with the risks being taken then you will see them redeeming in droves when large losses occur.

97. The other reason is that they trade too much, which is covered in chapter twelve, 'Speed and Size'.

Setting a volatility target

Imaginary conversation between a financial advisor and myself:

> Financial 'expert': "How much risk do you want to take?"
>
> Me: "What do you mean by risk?"
>
> Financial 'expert': "Er... well how would you define your tolerance for losing money?"
>
> Me: "Well it could be how much I'm prepared to lose next year. Or tomorrow. Or next week. Are you talking about the absolute maximum loss I can cope with, or the average, or the worst loss I'd expect 95 days out of 100 (the so called 'Value at Risk')? Which question would you like me to answer?"
>
> Financial 'expert': "Hold on. I need to speak to my supervisor..."

Joking aside, how do we answer this deceptively simple question? To keep things simple I use a single figure to measure appetite for risk – an expected **standard deviation**, which I call the **volatility target**. You can measure this as a percentage, or in cash terms, and over different time periods. So for example the **daily cash volatility target** is the average expected standard deviation of the daily portfolio returns. As it's a cash value you need to specify the currency that your account or fund is denominated in.

When I first discussed risk on page 39 I talked about **predictable** and **unpredictable risks**. Your volatility target is the long-term average of expected, predictable, risk. The exact predictable risk you have on any given day will depend on the strength of your **forecasts**, and on the current expected **correlation** of asset prices. You'll also face unpredictable risks if your forecast of volatility or correlations is wrong. In any case the actual amount you lose or gain on any given day will be random, since even a perfect estimation of risk only tells you what your *average* gains and losses will be.

I find it's easier to look at an **annualised cash volatility target**, which will be the annualised expected daily standard deviation of returns. As before you annualise by multiplying by the square root of time; given there are around 256 trading days in a year this will be 16. Beware: the annualised volatility target isn't the maximum, or even the average, you might expect to lose in a year.[98] Indeed it's quite probable you will sometimes lose more than that in a year.

98. There are several reasons why your expected average annual loss wouldn't be equal to the annualised expected daily standard deviation. Firstly, if your **Sharpe ratio** is greater than zero then your expected average annual loss will be smaller than one **sigma**. Secondly, in the event of losses you'd probably reduce your positions if you're using **trend following rules** or stop losses. Thirdly, as you'll see in the next chapter 'Position Sizing', you should reduce your positions when price volatility rises, which reduces losses in periods of rising risk. Also, as you'll see later in this chapter, you'd reduce your risk target throughout the year if you made losses, and vice versa. More technically if consecutive returns aren't independent, and have time series autocorrelation, then annualising by multiplying by 16 is a poor approximation.

It's also easier to separate out your cash account value and the appropriate level of risk to run on that money. The amount of cash you are trading with is your **trading capital**. You then decide what your volatility target will be as a percentage of that capital. If you multiply this **percentage volatility target** by your trading capital, then you'll get your volatility target in cash terms. So with a million dollars of trading capital and an annualised 10% percentage volatility target, you would have an annualised cash volatility target of $100,000.

In the rest of the chapter I'll be dealing with the implications of where you set your trading capital and percentage volatility target, rather than setting your cash volatility target directly. This means that amateur traders with £1,000 in their account can use the same guidelines to set percentage volatility target as multi-billion Euro hedge funds.

Here are the points to consider when setting your trading capital and percentage volatility targets:

1. **How much can you lose?**: How much money do you have to trade or invest?
2. **How much risk can you cope with?**: Can you afford to lose it all? Can you afford to lose half? What probability of losing half would you be comfortable with? What probability of losing 90% of it over ten years would make you lose sleep?
3. **Can you realise that risk?**: Given the instruments you are investing in and the safe amount of leverage (if any) you can use, can you actually hit the risk target?
4. **Is this level of risk right for your system?**: Given the characteristics of your trading system, expected **Sharpe ratio** and **skew**, does the amount of risk make sense?

How much can you lose?

The initial **trading capital** is the amount of cash you start with, bearing in mind that there is a chance that you might lose all or nearly all of it, although hopefully that's quite unlikely. I'll show you below how to set your **percentage volatility target** based on exactly how relaxed you are about losses.

For an institutional investor things are usually straightforward; you are given £100 million and you would use 100% of it. Sometimes you might not go 'all in' if you have guaranteed some of the capital, or need to retain cash for potential redemptions.

If you're investing your own money then your trading capital will depend on your savings and how much you are willing to commit to such a risky endeavour. In any case – and I can't emphasise this enough – never put in more than you can afford to lose. Never trade with borrowed money, or money earmarked to pay off debts.[99] Even if you follow the advice in this book to the letter there is still a remote chance that you will be unlucky enough to burn through virtually your entire account.

99. This isn't a ban on **leverage**, but if your trading capital is £100,000 you should actually have that available to lose, and none of it should be borrowed. I'll discuss leverage in more detail below.

Can you cope with the risk?

Let's say you decide to put $100,000 of **trading capital** into your account and run with a 200% **volatility target**; equating to an **annualised cash volatility target** of $200,000. Could you cope with losing $20,000 in one day? What about having a cumulative loss, or draw-down, of over $60,000? If it isn't your money, could your client cope with it?

I hope so because you're likely to see those kinds of losses within the first few weeks of trading! As table 20 shows, a $20,000 loss would typically be seen every month, and a $62,000 cumulative loss around 10% of the time.[100]

TABLE 20: WHAT KIND OF LOSSES DO WE SEE FOR A PARTICULAR VOLATILITY TARGET?

	Percentage volatility target			
Expected	**25%**	**50%**	**100%**	**200%**
Worst daily loss each month	$2,500	$5,000	$10,000	$20,000
Worst weekly loss each year	$6,900	$14,000	$28,000	$55,000
Worst monthly loss every ten years	$16,000	$32,000	$63,000	$80,000
Worst daily loss every 30 years	$5,400	$11,000	$22,000	$43,000
10% of the time, the cumulative loss will be at least	$9,300	$15,000	$30,500	$62,000
1% of the time, the cumulative loss will be at least	$11,000	$18,500	$37,000	$75,000

The table shows various expected losses (rows), and different percentage volatility targets (columns), given trading capital of $100,000, assuming Sharpe ratio 0.5 and zero skew with Gaussian normal returns.

Now, table 20 assumes you have zero **skew** in your returns. Are you running a positive skew **trend following** system, or a negative skew **relative value** or **carry** rule? Systems with different **skew** have varying risk properties. As table 21 shows, the worst days, weeks and months for a negative skew strategy are much nastier than with zero skew. For positive

100. I've assumed a **Sharpe ratio (SR)** of 0.5 here, and the returns are drawn from a **Gaussian normal distribution**. Using a different SR won't affect these numbers much, although the annual loss figures will be slightly better (worse) if the true Sharpe is higher (lower).

skew strategies large losses are much less likely, as you can see in table 22. However the typical cumulative loss is higher.

With negative skew it's vital to have sufficient capital to cope with the very bad days, weeks and months you will occasionally see. This is especially true with high leverage and the risk your broker will make a margin call at the worst possible time. With positive skew the difficulty is psychological; committing to a system when you spend most of your time suffering cumulative losses.

TABLE 21: HOW DO TYPICAL LOSSES LOOK WITH NEGATIVE SKEW? INDIVIDUAL LOSSES ARE HIGHER THAN IN TABLE 20, BUT CUMULATIVE LOSSES ARE SMALLER

	Percentage volatility target	
Expected	**25%**	**50%**
Worst daily loss each month	$3,100	$6,100
Worst weekly loss each year	$8,500	$17,000
Worst monthly loss every ten years	$18,100	$36,000
Worst daily loss every 30 years	$11,500	$23,000
10% of the time, the cumulative loss will be at least	$3,700	$7,400
1% of the time, the cumulative loss will be at least	$7,100	$14,000

The table shows various expected losses (rows) and different percentage volatility targets (columns), for a negative skew option selling strategy given trading capital of $100,000.[101] *The strategy has a Sharpe ratio of 0.5 and skew of around -2.*

101. Results are from a **back-test** of a strategy which involves persistently selling futures option straddles.

TABLE 22: HOW DO LOSS PATTERNS CHANGE FOR POSITIVE SKEW? WITH POSITIVE SKEW INDIVIDUAL LOSSES ARE MUCH BETTER THAN IN TABLES 20 AND 21, BUT AVERAGE CUMULATIVE LOSSES A LITTLE WORSE

	Percentage volatility target	
Expected	**25%**	**50%**
Worst daily loss each month	$2,000	$4,000
Worst weekly loss each year	$6,100	$12,000
Worst monthly loss every ten years	$15,000	$30,000
Worst daily loss every 30 years	$2,800	$5,700
10% of the time, the cumulative loss will be at least	$11,000	$22,000
1% of the time, the cumulative loss will be at least	$12,000	$24,000

This table shows various expected losses (rows) given different percentage volatility targets (columns), for a positive skew trend following strategy, with trading capital of $100,000.[102] *The strategy has a Sharpe ratio of 0.5 and skew of around 1.0.*

Earlier I said you could frame risk by how much you are prepared to lose in a lifetime. That's difficult to quantify without knowing your life expectancy, so let's assume you will trade for ten years. In table 23 I show the chances of ending a decade-long trading career with less than half, and less than 10%, of your **trading capital** left, given a certain percentage target volatility.

Only you know how much risk you can cope with. However you, or your clients, must be able to stomach the likely losses involved. If you can't then you should set a lower percentage volatility target, or if possible consider using less initial trading capital.

102. Figures from **back-testing** a futures system using a mixture of relatively fast **trend following rule variations**.

TABLE 23: WHAT ARE THE CHANCES OF LOSING ALL, OR MOST, OF MY MONEY?

	Percentage volatility target			
	25%	50%	100%	200%
Chance of losing half	<1%	10%	40%	93%
Chance of losing 90%	<1%	1.1%	22%	88%

The table shows chances of ending a ten-year trading career losing a given proportion (rows), of initial trading capital given different percentage volatility targets, assuming Sharpe ratio is 0.50 and zero skew.[103]

Can you realise that risk?

If you're investing in leveraged derivatives like futures and spread bets then very high levels of risk are attainable, even if they aren't desirable. Such systems can easily run at over 100% annualised **target volatility** with margin to spare.

But if you can't get enough, or any, leverage then you might have a problem achieving your target volatility. If you are buying short-term government bonds with an expected volatility of perhaps 5% a year, then without leverage it's impossible to create a portfolio with a 50% volatility target. With no leverage you are restricted to the amount of *natural* risk that your **instruments** have. With only 100% leverage you are limited to twice that natural risk, and so on.

Because it's mostly a problem for non-leveraged **asset allocating investors** I'll go into more detail about this in the relevant part of part four, chapter fourteen.

Is there too much leverage?

Even if you are able to leverage up as required to hit a particular **percentage volatility target**, it would be very unwise if excessive gearing is needed. This is particularly problematic for negative **skew** instruments and trading strategies, which tend to have low natural risk – until they blow up.

I've mentioned before the huge appreciation of the Swiss franc that happened in just minutes in January 2015. At the start of the day in question the natural risk of holding a position in EUR/CHF was tiny, at around 1% a year. If this was the only instrument you were trading then to achieve a 50% annualised volatility target would have needed

103. The results would be slightly worse for negative **skew**, and slightly better for positive skew. For example with a 200% **volatility target** the chances of losing half your capital over ten years are 90% with positive skew of 1.0 and 97% with negative skew of -2.0.

50 times leverage. Retail FX brokers had no compunction in allowing this, with leverage up to 500 times available from some providers.

If you had been on the wrong side of this move, with your entire trading capital leveraged 50 times, then a 2% appreciation would have wiped you out. But the actual move was over 16%! Only those with leverage of 7 times or less would have survived the day, which implies a maximum achievable 7% volatility target.

You should ensure that with a given percentage volatility target any individual position would not wipe you out after the largest conceivable move. Diversifying amongst many different **instruments** will also help, and we'll return to that in chapter eleven, 'Portfolios'. A 16% move with 50 times leverage would have been just about survivable if EUR/CHF was only 10% of your portfolio, assuming no other losses had occured elsewhere.

Ideally such low **volatility** instruments, requiring insanely high leverage, should be excluded from any trading system.

Is this the right level of risk?

Suppose you've decided on a 200% **volatility target**. You've got the leverage you need; but you haven't got carried away. Furthermore you're confident that you will cope with the spectacularly bumpy ride tables 20 to 23 imply you'll be getting. Assuming you are a profitable trader, should you then set your target at 200% and expect to end up incredibly wealthy through the magic of compound interest?

The short answer is no. There is a Goldilocks level of risk – not too little and not too much. Even if you are willing and able to go above this level you shouldn't, as you will end up with more than your tongue getting burnt.

Naively if you expect to be profitable then you should bet as much as you're allowed to. However this ignores the *compounding* of returns over time. Suppose you have a fantastic expected average return of 100% on each trade for a given bet size. You then lose 90% of your capital on your first trade and make 190% on your next trade. Unfortunately there is only 29% of your cash left, even though you've achieved the expected average return of 100% per trade.[104] To maximise your final profits, the optimal bet to make is actually a quarter of the original size.

Nearly all professional gamblers, many professional money managers and some amateurs in both fields know that this optimal point should be calculated using something called the **Kelly criterion**.[105] Kelly has some useful but potentially dangerous implications for how you should set your **percentage volatility target**.

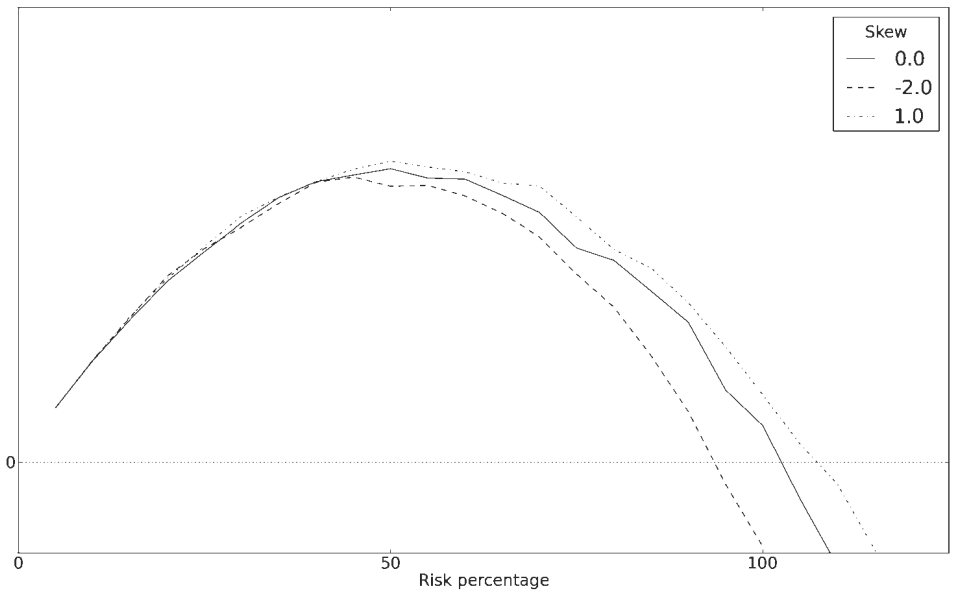
104. After the first loss you have 10% of your capital left. A 190% return on this generates another 19%, for a total of 29%.

105. The Kelly criterion maximises the geometric mean of returns, which is what we get from compounding. It was invented by physicist John Kelly in 1956. There is an excellent and highly readable book on this subject – *Fortune's Formula* by William Poundstone.

A simple formula can be used to determine how you should set your volatility target, given the underlying **Sharpe ratio (SR)** of your trading system. You should set your volatility target at the same level as your expected SR. So if you think your annualised SR will be 0.25 then you should have a 25% annualised volatility target.

You can see this in figure 21, where for an SR 0.5 system the best performance is achieved with the optimal 50% volatility target. This is true for all three systems shown, regardless of **skew**.

FIGURE 21: KELLY CRITERION IMPLIES THE OPTIMAL RISK PERCENTAGE FOR A SHARPE RATIO 0.5 SYSTEM IS 50%. WITH HIGHER RISK THINGS GO BADLY, ESPECIALLY FOR NEGATIVE SKEW STRATEGIES.



The X-axis shows the percentage volatility target and the Y-axis the geometric mean of returns which Kelly optimises.

TABLE 24: BIG SHARPE MEANS BIGGER RISK AND EXPONENTIALLY BIGGER PROFITS

Expected Sharpe ratio	Optimum percentage volatility target	Expected return
0.2	20%	4%
0.5	50%	25%
0.75	75%	56%
1.0	100%	100%
2.0	200%	400%

The table shows the expected annual return (as a percentage of trading capital), given different Sharpe ratio (SR) (rows) and using the optimal Kelly percentage volatility target. Expected return is SR multiplied by percentage volatility target.

This finding is potentially dangerous when used by an over confident investor. It's very easy with **back-testing** software to get **over-fitted** performance with a Sharpe ratio (SR) of 2, 3 or even higher. If you believe those are attainable then a risk percentage of 100% or 200% seems justified. As table 24 shows, running at a 200% risk with SR of 2.0 implies huge returns of 400% a year!

Unfortunately many people with capital of $20,000 will conclude it's possible to earn 400% a year, or $80,000, as full-time traders. There are also plenty of brokers who will happily provide them with the necessary leverage. Most of these people will quickly lose their $20,000, as they won't achieve their expected SR. It's very difficult to know exactly what your true Sharpe ratio really would have been in the past, with back-tests giving you only a rough upwardly estimate, and it's utterly impossible to know what SR to expect in the future.

Even if you had a crystal ball, and knew your expected Sharpe precisely, you could be unlucky and have a decade or more of sub-par performance. Figure 21 shows that if you realise an SR of only 0.5 then a 200% volatility target will see you ending up deep underwater. In general if you get your estimate of SR wrong and bet more than the optimal then you have a high chance of losing your shirt.

Recommended percentage volatility targets

I run a highly diversified futures trading system with around 45 **instruments**, eight **trading rules** drawn from four different styles, and 30 **trading rule variations**. In a 35 year **back-test**, conservatively fitted with **out of sample bootstrapping**, it has a **Sharpe ratio (SR)** of around 1.0 after costs, but the highest **volatility target** I'd advocate using

for it is 37%, rather than the 100% suggested by the **Kelly criterion** and the back-tested SR.[106] Why such a conservative number – am I a wimp?

There are several reasons for my caution. Firstly, it's unlikely a back-tested SR will be achieved in the future. On average realised performance is never as good as in back-tests. This is because it's very easy to over-fit if you ignore the advice in chapters three and four. Additionally it's difficult with **ideas first** testing to avoid using only trading rules that you already know would have worked.

Even if you could avoid over-fitting actual profits are unlikely to be as high as they could have been in the past. This is because future asset returns are likely to be lower than in the historical data set we usually use for back-testing, as I discussed in chapter two, 'Systematic Trading Rules', in the section on achievable Sharpe ratios (from page 46).

To find realistic achievable SR from back-test results a good rule of thumb is to use the ratios in table 14 (page 90). These suggest that for an out of sample bootstrap, as I've used in my own system, a ratio of 0.75 should be applied to find a more realistic Sharpe ratio. Much lower ratios should be used if you haven't been as careful with your fitting. I also said in chapter two that I think the absolute maximum SR that **staunch systems traders** should expect to achieve is 1.0, regardless of how good their back-test is.

Secondly, using the full Kelly criteria is far too aggressive, because of the risk of getting a poor run of luck and the large drawdowns that can result, even if SR expectations are correct.[107] In table 23 someone using the correct Kelly target of 50% would have a 10% chance of losing half their money after ten years; which most people would find worrying. It's far better to use **Half-Kelly** and set your risk at half the optimal. Column A of table 25 shows the recommended percentage volatility target for a given realistic back-tested SR.

For my own system I started with the back-tested Sharpe ratio of 1.0. Multiplying by 0.75 (as I'm using out of sample bootstrapping) from table 14, this gives me a realistic SR of 0.75. With full Kelly criterion betting that would be a 75% volatility target, which I then halved to get 37% (rounding down).

This assumes your trading system, like mine, has zero or positive **skew**. You should be very careful if you have expected negative skew. As figure 21 shows, the penalty for too much risk is greater with negative skew than when skew is positive, or zero. As I discussed in chapter two, many negative skew strategies have fantastic SR in back-test, but I advise you to run them at half the risk you'd use for a more benign trading system. Column B of table 25 shows the recommended percentage volatility target for negative skew systems.

106. At the time of writing I'm using a 25% **volatility target**, reflecting my relatively low appetite for risk.
107. Here is famous Kelly fan, hedge fund manager and Blackjack card counter Ed Thorpe: "My experience has been that most cautious … investors who use Kelly find the frequency of substantial bankroll reductions to be uncomfortably large." (Quoted in *Fortune's Formula*.)

TABLE 25: WHAT VOLATILITY TARGET SHOULD STAUNCH SYSTEMS TRADERS USE?

	Recommended percentage volatility target	
Realistic back-tested SR	(A) Skew>0	(B) Negative skew
0.25	12%	6%
0.40	20%	10%
0.50	25%	12%
0.75	37%	19%
1.0 or more	50%	25%

The table shows the recommended percentage volatility target for those who can back-test their dynamic trading systems, depending on the skew of returns (columns) and achievable back-tested Sharpe ratios (SR) (rows) after making adjustments to simulated results from table 14 on page 90. Optimal volatility target is calculated using Half-Kelly. For negative skew strategies this is cut in half again. We assume a maximum SR of 1.0 is achievable.

The returns of **asset allocating investors** are limited by their use of a **static** trading strategy. With a small, relatively undiversified portfolio you shouldn't expect high Sharpe ratios. As I said in chapter two, if you're holding a dozen equities in different industries but from the same country then you probably will achieve an SR of around 0.20. Those with larger portfolios diversified across multiple **asset classes** could get a maximum realistic SR of 0.4.

Column C of table 26 shows the correct targets given asset allocators' SR expectations. However, as you'll see in chapter fourteen, it's unlikely even this level of volatility can be achieved as these investors don't use leverage.

Semi-automatic traders have systems which cannot be back-tested and usually have a small, relatively un-diversified, set of ad-hoc instruments. I would initially assume an achievable SR of 0.20 unless you are very experienced, are trading across multiple asset classes, and have a good track record with real money. As you saw in chapter two, in my opinion an SR of 0.5 is the maximum safe achievable level, so you should set the volatility target at no more than 25%.

Again the target should be halved if you are trading a strategy that is likely to have negative skew, such as selling option volatility, or exhibits a similar return pattern with steady profits on most bets with occasional large losses. Columns D and E of table 26 show the appropriate volatility targets for this type of trader.

TABLE 26: WHAT VOLATILITY TARGET SHOULD ASSET ALLOCATING INVESTORS AND SEMI-AUTOMATIC TRADERS USE?

	Recommended percentage volatility target		
Expected SR	**(C) Asset allocating investor**	**(D) Semi-automatic trader, zero or positive skew**	**(E) Semi-automatic trader, negative skew**
0.20	10%	10%	5%
0.30	15%	15%	7%
0.40	20%	20%	10%
0.5 or more	20%	25%	12%

This table shows the recommended percentage volatility target depending on the type of trader and expected skew (columns), and Sharpe ratio (SR) expectations (rows). The optimal volatility target is calculated using Half-Kelly. We assume asset allocators shouldn't expect more than 0.40 SR and semi-automatic traders won't get more than 0.50 SR. Asset allocators are assumed to have zero skew. We halve volatility targets for negative skew semi-automatic trading.

This implies that nobody will use more than a maximum 50% volatility target and most people should use less. Tables 20 to 23 illustrate that a 50% annualised volatility will mean some pretty substantial losses from time to time. Just because the volatility targets in tables 25 and 26 are optimal it doesn't mean they will suit you, or that your broker will permit them. It just means you should *never* run a higher risk target than this.

When the percentage volatility target should be changed

I don't advocate changing your **percentage volatility target** since there is a potential risk of *meddling*; reducing it when you don't trust your system and increasing it when it agrees with you. The exception is if you have grossly miscalculated your tolerance for risk. You begin trading on day one with a 50% target, but the first big loss then sends you or your investors into a panic. In this case you should significantly reduce your percentage target. However it should ideally be a one off change and only ever downwards.

To avoid this scenario it's better to start with significantly lower **trading capital** and gradually increase it until you have the full amount invested. Keep the percentage volatility target fixed and allow your cash volatility to increase up to the point just before you get uncomfortable. This also helps with gaining confidence in your trading strategy and testing any automated systems.

Rolling up profits and losses

Once set your **percentage volatility target** shouldn't need changing. However your **trading capital** will definitely change from its initial value. This implies that your **cash volatility target** will also be adjusted over time.

Let's imagine you have trading capital of $100,000 and after reading this chapter you've decided on a 30% volatility target. You begin trading with the appropriate $30,000 **annualised cash volatility target**. Then on day one you lose $2,000. Should you continue to use a $30,000 volatility target?

You should not. The implication of the **Kelly criterion** is that you should adjust your risk according to your current capital. You did have $100,000 of capital, but now you only have $98,000. Instead of having a 30% volatility target ($30,000 of $100,000) you effectively have a target of $30,000 divided by $98,000 = 30.6%. This might not seem a big deal, but you are now betting more than you intended and you've slightly increased your chances of going bankrupt. If you make further losses the situation could deteriorate fast.

The correct thing to do is to reduce your volatility target to 30% of your current capital of $98,000, or $29,400.

Now suppose things start to improve and you make $2,000 back. Your accumulated losses have reduced to zero and your volatility target will be back to 30% of $100,000, or $30,000. After another profitable day making $3,000 you're now in positive territory with an account value of $103,000 – higher than your starting capital. Should you now increase your risk appetite above its initial level?

If you want to maximise your wealth then Kelly says you should roll up your profits and increase your capital.[108] The new volatility target will be 30% of $103,000, or $30,900. This means you will be compounding your returns, which over the long run will increase them faster.

You should also reduce your trading capital, and hence your cash volatility target, if you withdraw money from your trading account or investors redeem. Similarly if you put more funds in then you would normally increase your maximum capital. There are exceptions such as if you are putting cash into a leveraged account to meet margin or reduce borrowing, but you don't want to increase the amount of capital at risk.

In my own trading an automated process checks my account value, and adjusts risk accordingly, on an hourly basis. If your system is not automated and if you are running with more than a 15% volatility target I would recommend checking at least daily. With a lower target you can check more infrequently, but if you're using leverage I'd advise always calculating your volatility target at least once a week.

108. Someone who is trading as a hobby and starting with a small stake would probably do this. Those investing other people's money should also add all profits after fees to their trading capital, so that returns are compounded, unless they are committed to paying coupons or guaranteeing a certain proportion of the initial investment made. If you are more risk averse then rolling up profits might not make sense, if for example you live partly or wholly off your trading income. This is the approach that I take.

What percentage of capital per trade?

Traditional money management systems allocate a certain percentage of **trading capital** to be risked on each trade or bet. If you're familiar with these systems you might be wondering how this relates to the volatility targeting done here. It is possible to infer the **percentage volatility target** that is implied for a particular trading system if you know the approximate holding period, the average number of positions held and the maximum amount of capital put at risk on each trade. You just need to assume that the average bet is half the maximum, which is the same ratio between my recommended average forecast of 10 and maximum of 20.

Table 27 gives the results, assuming an average of two positions are held at once. If a greater or fewer number are traded on average, then you just need to multiply or divide the figures appropriately. So with an average of four positions you'd double them, and for a single position halve them.

TABLE 27: WHAT IS THE VOLATILITY IMPLIED BY A TRADING SYSTEM'S HOLDING PERIOD AND BET SIZE?

	Implied percentage volatility target				
	Maximum percentage of capital at risk per trade				
Average holding period	**1%**	**2.5%**	**5%**	**10%**	**20%**
1 day	40%	100%	200%	!	!
1 week	16%	40%	80%	160%	!
2 weeks	8%	19%	38%	76%	152%
6 weeks	4%	10%	21%	41%	82%
3 months	3%	7%	13%	27%	53%

The table shows the implied annualised percentage volatility target for a given average holding period (rows) and maximum percentage of capital allocated to each bet (columns), assuming an average of two bets are held in the portfolio. "!" indicates volatility greater than 200% per year.

As an example take the system which I briefly discussed in the introductory chapter. This held positions for around a week, with no more than 10% of capital at risk. Let's assume on average that two bets are made at once, although the author wasn't clear on this point (a common shortcoming in trading books).

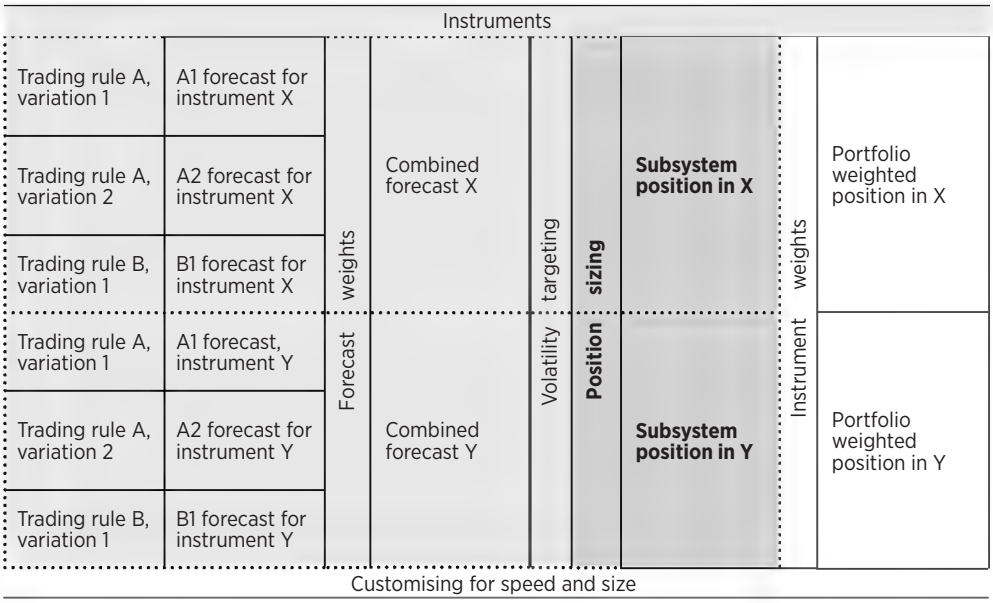
This all sounds fairly sedate but from the table this works out to a suicidal 160% volatility target. If this target is **Kelly optimal** then the achievable **Sharpe ratio** must be at least 1.6, implying an expected return of 1.6 × 160% or 256% a year! There is some serious overconfidence at work here. Worse still, this is nowhere **near** the most aggressive system I've ever seen.

Summary of volatility targeting

Percentage volatility target	Desired long run expectation of annualised **standard deviation** of percentage portfolio returns. A maximum of: • The level of risk that you are comfortable with given tables 20 to 23. • What is practically attainable given your access to **leverage** and the natural risk of your **instruments** (see the **asset allocating investor** example in part four for more details). • The highest level that is safe given the natural volatility of your instruments, and how much leverage they need. Avoid very low **volatility** instruments requiring insanely high leverage. • The recommended percentage volatility in tables 25 and 26, depending on the **back-tested** or expected **Sharpe ratio** and **skew** of your trading system, and the type of trader or investor you are. Percentage volatility target should remain unchanged.
Trading capital	The amount of capital you currently have at risk in your account. Every day you should add profits and any injections of funds. Deduct any losses and withdrawals made from the account. Measured in currency.
Annualised cash volatility target	Long run expectation of annualised **standard deviation** of daily portfolio returns. Equal to the **trading capital** multiplied by the **percentage volatility target**. Measured in currency.
Daily cash volatility target	Long run expectation of daily standard deviation of daily portfolio returns. **Annualised cash volatility target** divided by 'the square root of time', which for a 256 business day year is 16. Measured in currency.

In the next chapter I will come back to the final piece of advice I got from Sergei and think about how the risk of an instrument determines what size position we should take.

Chapter Ten. Position Sizing



LET'S COME BACK TO SERGEI'S THREE QUESTIONS FROM CHAPTER seven. First, how much do you like this trade? This is the **forecast** for each instrument. Secondly, how much can you afford to lose? You should know how many chips of **trading capital** you're willing to put down on the hypothetical casino baize depending on your desired **target volatility**.

So far, so good. But how many shares, bonds, spread bet points or futures contracts should you actually buy or sell? What does a **combined forecast** of -6 and a £1,000,000 **annualised cash target volatility** mean in practice if you're trading crude oil futures in New York? To answer this we need to come back to Sergei's third question: how risky is your trade?

Once you know this you can move from the abstract world of forecasts and trading rules to deciding the size of actual positions in real financial **instruments**.

Chapter overview

How risky is it?	If you own one unit of an instrument how much risk does that expose you to?
Volatility target and position risk	What is the relationship between your cash volatility target and the risk of each instrument?
From forecast to position	Given how confident you are in your forecasts, the cash volatility target you have and the position risk, what size position should you be holding?

How risky is it?

What is the expected risk of holding an **instrument**?[109] If you own one Apple share or one crude oil futures contract, how much danger are you exposed to?

Position block and block value

Let's start by asking a philosophical question. What exactly is 'one' of an instrument? I define this as the **instrument block**. If 'one' of an instrument goes up in price by 1% how much do you gain or lose? This is the **block value**.

These definitions will seem trivial to equity investors. 'One' Apple share is exactly that – one Apple share. If a share has a price of $400 it will cost exactly $400 to buy one block. If the price goes up by 1% from $400 to $404 then you will gain $4. Apple shares, and most other equities, have a natural **instrument block** of one share and a block value of 1% of the price. Sometimes though to reduce costs you'll trade in larger blocks, usually of 100 shares. In this case the block value will be 100 × 1% × share price, which is equal to the cost of one share.

But life isn't always that simple. For example you can use a UK financial spread betting firm to bet on the FTSE falling at £10 a point. If the FTSE rises 1% from 6500 to 6565 you would lose £650. Here the block value is ten times 1% of the price.

Futures contracts also have non-trivial block values. WTI crude oil futures on NYMEX are quoted in dollars per barrel. But each futures contract is for 1000 barrels. This means

109. Notice we're dealing here with expected, **predictable risk**, as discussed on page 39.